Vo Thanh Nguyen

AI Engineer

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Portfolio | GitHub | LinkedIn

Objective

AI Engineer and Researcher with 2+ years of engineering and research experience in Machine Learning, Computer Vision, World Models, and Self-Driving Systems. Proven track record of bridging applied deep learning research with production-grade software, from multimodal perception and trajectory planning to real-time edge deployment. Eager to engineer scalable, high-impact AI solutions for complex real-world environments.

Technical Skills

Programming & Frameworks: Python, C++, SQL, Bash, PyTorch, TensorFlow, CUDA, OpenCV, Hugging Face, Scikit-learn

Computer Vision Models: YOLO, RF-DETR, Mask2Former, SAM 2, DINOv2/v3, CLIP, V-JEPA, Qwen-VL, Video Diffusion

Core Vision Tasks: Object Detection, Multi-Object Tracking, Semantic/Instance Segmentation, Trajectory Planning, VLM

Edge & Systems: TensorRT (FP16/INT8), ONNX Runtime, OpenVINO, High-Throughput IPC, NVIDIA Jetson, Docker

Professional Experience

- **AI Researcher / Engineer** Dec 2024 – Present

  HCM-UTE Research Lab

  - **Autonomous Vehicle Software Stack (BFMC)** – *Software Lead*:
    - Architected an end-to-end autonomous driving stack on NVIDIA Jetson Orin Nano, integrating multi-sensor fusion and high-throughput, low-latency inter-process communication (IPC) for concurrent real-time execution.
    - Developed a unified multi-task perception and planning pipeline for object detection, multi-object tracking, lane segmentation, and trajectory prediction, quantized with TensorRT INT8 ($>$100 FPS, $<$25ms latency).

      Achievement: Global Top 9 (Romania Finals 2025) and **1st Nationally / 150M VND Bosch Grant** (2026).
  - **Sim-to-Real Traffic Scene Understanding** – *Co-First Author*:
    - Leveraged latent spatiotemporal representations from the V-JEPA world model to bridge the sim-to-real domain gap for traffic video question answering and scene captioning.
    - Combined structured visual fact extraction with Vision-Language Models (VLMs) and relational consistency reasoning to reduce hallucinations in complex traffic events.

      Achievement: 1st Worldwide at the NVIDIA AI City Challenge 2026 Track 2 (ECCV 2026 Workshop Oral).
  - **Generative Traffic Forecasting** – *Co-First Author*:
    - Exploited V-JEPA latent spatiotemporal representations to capture multi-agent motion dynamics and condition video diffusion models for future traffic scene generation.
    - Aligned predictive world-model priors with diffusion synthesis and temporal refinement to preserve physical plausibility and visual coherence across generated frames.

      Achievement: Rank 3 Worldwide at the NVIDIA AI City Challenge 2026 Track 5 (ECCV 2026 Workshop).

  Tech Stack: Python, C++, PyTorch, CUDA, TensorRT, YOLO, RF-DETR, Mask2Former, SAM 2, V-JEPA, VLMs, Video Diffusion, NVIDIA Jetson Orin, Docker.

- **AI Automation Engineer** Aug 2025 – Nov 2025

  Learning Chain

  - Built automated data ingestion pipelines for multi-source web and social media crawling with real-time deduplication.
  - Automated end-to-end LLM content generation and multi-channel social media publishing with SEO optimization.
  - Implemented vector-based RAG pipelines for document retrieval and automated scheduled workflow versioning.

  Tech Stack: Python, n8n, LLMs, Vector Databases, RAG, RESTful APIs, Webhooks, Git.

- **AI Engineer** Mar 2024 – Feb 2025

  AIoT & Semiconductor

  - Deployed an industrial vision pipeline for automated PCB defect detection and segmentation, achieving 98.2% accuracy.
  - Built a real-time driver monitoring system fusing 478-point face mesh, gaze vectors, and 3D head pose estimation.
  - Engineered a high-throughput, low-latency IPC pipeline across TensorRT and ONNX backends (3.2$\times$ throughput).

  Tech Stack: Python, OpenCV, YOLO, TensorRT, OpenVINO, ONNX Runtime, MediaPipe, Hailo NPU, Docker.

Publications

• **Sim-to-Real Traffic Scene Understanding by Decoupling Semantics from Caption Generation with V-JEPA**
Nguyen Hoai Thuong Bui*, **Thanh Nguyen Vo***, Trinh Tra Giang Nguyen*, Ha Duc Bui†
ECCV 2026 Workshop (Oral Presentation) **[arXiv]** May 2026

• **JEPA Guided Diffusion: Predictive Vision-Language Conditioning for Generative Traffic Forecasting**
Trinh Tra Giang Nguyen*, **Thanh Nguyen Vo***, Nguyen Hoai Thuong Bui*, Ha Duc Bui†
ECCV 2026 Workshop (Poster Presentation) **[arXiv]** May 2026

• **An Integrated Perception-to-Planning Framework for Autonomous Driving using YOLOv11 and Frenet**
Thanh Nguyen Vo*, Trinh Tra Giang Nguyen*, Lu Anh Khoi Le, Ha Duc Bui†
IEEE GTSD 2026 Proceedings **[IEEE Xplore]** June 2026

• **LENS: Likelihood Evaluation and Neighbor Smoothing for Attribute-Based Person Retrieval on Unseen Cameras**
Nguyen Dang Duy Tan, **Vo Thanh Nguyen**, Linh T. T. Tran, Huynh Nguyen Phat
IEEE/CVF WACV 2027 Workshop (Accepted) **[arXiv]** Oct 2026

* Equal contribution / Co-first authors † Corresponding author

Education

Ho Chi Minh City University of Technology and Engineering (HCMUTE) 2022 – 2026
Bachelor of Engineering in Robotics and Artificial Intelligence GPA: 3.28 / 4.0 | **TOEIC: 735 / 990**

Relevant Coursework: Artificial Neural Networks: 9.6/10, Computer Vision: 9.5/10, Artificial Intelligence: 9.1/10.

- **Graduation Thesis (Grade: 9.2/10):** *End-to-End Control System for Autonomous Vehicles*.

Honors & Awards

AI, Autonomous Systems & Data Competitions:

- **Champion (1st Worldwide, Track 2)** – NVIDIA AI City Challenge [Cert] 2026
- **Top 3 Worldwide (Track 5)** – NVIDIA AI City Challenge [Rank] 2026
- **Top 9 Finalist Worldwide (Best in Asia)** – BFMC Romania [Cert] 2025
- **150M VND Bosch Sponsorship** (Team "Drive Beyond") – BFMC [Grant] 2026
- **Top 5 Finalist** – Datathon 2026: The Gridbreakers (VinUniversity) [Cert] 2026

Competitive Programming & Algorithms:

- **Third Prize** – ICPC Vietnam National Programming Contest [Cert] 2024
- **Bronze Medal** – Da Nang Code League [Cert] 2024
- **Third Prize** – Mastering IT HCMUTE [Cert] 2023
- **Second Prize (Specialized Division)** – OLP Vietnam [Cert] 2023
- **Second Prize (Non-Specialized Division)** – OLP Vietnam [Cert] 2022

Academic Honors:

- **Talent Scholarship** – HCMUTE Academic Merit Award [Cert] 2022 – 2023

References

Dr. Ha Duc Bui – Head of Research Lab, HCMUTE buihaduc@hcmute.edu.vn